

Maximal violation without maximal global randomness in a cyclic Bell family

Exact counterexamples in every dimension $d \geq 4$

Alec Kriebel

with heavy assistance from ChatGPT 5.6 Sol

26 July 2026

Abstract

Perito, D’Avino, Jung, Mironowicz, Acin, and Augusiak conjectured that maximal violation of an augmented cyclic Bell functional certifies $2 \log_2 d$ bits of global randomness. We disprove the maximal-randomness assertion for every $d \geq 4$. For each such d , a sparse weighted-shift strategy on $\mathbb{C}^d \otimes \mathbb{C}^d$, with no adversarial side information, attains the exact Bell maximum

$$2 \csc\left(\frac{\pi}{2d}\right) + 1$$

while the designated joint output distribution is nonuniform. Quantitatively,

$$G(AB \mid 1, d, E) \geq \frac{1}{d^2} + \frac{2 \sin(\pi/d) \sin(3\pi/d)}{d^2(d-1)} > \frac{1}{d^2}.$$

Both one-party marginals remain uniform, so the obstruction concerns global joint randomness rather than a local-output bias. At $d = 4$, a compact certificate over $\mathbb{Q}(\zeta_{16})$ gives probabilities $1/32$ and $3/32$, hence $G = 3/32 > 1/16$. We reproduce the exact Bell upper certificate, prove the infinite counterexample family from first principles, and provide an exact-arithmetic verifier for the finite witness.

Status. This is an unrefereed, AI-assisted research note. Its proof and exact certificate are public for audit but have not received independent expert review.

1 Definitions and result

Fix $d \geq 2$ and $\omega = e^{2\pi i/d}$. Alice has order- d unitaries A_0, A_1 , Bob has order- d unitaries $B_0, \dots, B_d$, and

$$\mathcal{I}_d = \frac{1}{2} \sum_{x=0}^1 \sum_{y=0}^{d-1} \left(\omega^{xy} A_x \otimes B_y + \omega^{-xy} A_x^\dagger \otimes B_y^\dagger \right), \quad (1)$$

$$\bar{\mathcal{I}}_d = \mathcal{I}_d + \frac{1}{2} \left(A_0 \otimes B_d + A_0^\dagger \otimes B_d^\dagger \right). \quad (2)$$

These are Eqs. (11) and (10), respectively, of Ref. [1]. Put

$$M_d = 2 \csc\left(\frac{\pi}{2d}\right). \quad (3)$$

The exact operator bound $\mathcal{I}_d \leq M_d \mathbf{1}$, proved in a companion note [2], gives

$$\bar{\mathcal{I}}_d \leq (M_d + 1) \mathbf{1}. \quad (4)$$

Section 2 recalls a self-contained proof.

For the output settings $x = 1, y = d$, let σ_E^{ab} be Eve's subnormalized conditional states. Their guessing probability is

$$G(AB \mid 1, d, E) = \max_{\{Q_{ab}\}} \sum_{a,b} \text{Tr}(Q_{ab} \sigma_E^{ab}).$$

Maximal private global randomness would require $\sigma_E^{ab} = \rho_E/d^2$, and hence $G = 1/d^2$.

Theorem 1.1 (Failure of maximal global randomness). *For every integer $d \geq 4$, there is an explicit admissible strategy on $\mathbb{C}^d \otimes \mathbb{C}^d$, with $\mathcal{H}_E = \mathbb{C}$, such that*

$$\langle \bar{\mathcal{I}}_d \rangle = M_d + 1$$

but

$$G(AB \mid 1, d, E) \geq \frac{1}{d^2} + \frac{2 \sin(\pi/d) \sin(3\pi/d)}{d^2(d-1)} > \frac{1}{d^2}. \quad (5)$$

For the same strategy, each Alice and Bob marginal is uniform. Thus maximal violation does not force maximal observed joint uniformity, and therefore cannot force maximal private global randomness.

Under the operator definitions (1)–(2), the maximum in Theorem 1.1 is fixed by (4) and an attaining strategy. Equation (17) of Ref. [1] likewise gives M_d for the unaugmented strategy, and its $d = 3$ discussion gives the augmented value $5 = M_3 + 1$. The sentence preceding Conjecture 2 and the conjecture itself print an additional factor d in the first term. We therefore treat the displayed operators and Eq. (17) as authoritative. The counterexample addresses the conjecture's substantive assertion that *maximal violation* of the first augmented family certifies $2 \log_2 d$ random bits.

2 The exact Bell upper certificate

We reproduce the argument because global optimality is essential to the counterexample. Write

$$C_y = A_0 + \omega^y A_1, \quad U = A_0^\dagger A_1, \quad F_d(U) = \sum_{y=0}^{d-1} |\mathbf{1} + \omega^y U|.$$

Lemma 2.1 (Polar positive-factor identity). *Let $C = V|C|$ be the polar decomposition of a finite-dimensional operator; V may be a partial isometry. If B is unitary, then*

$$P = |C^\dagger|^{1/2} \otimes \mathbf{1} - V|C|^{1/2} \otimes B$$

satisfies

$$\frac{|C| + |C^\dagger|}{2} \otimes \mathbf{1} - \text{Re}(C \otimes B) = \frac{1}{2} P^\dagger P. \quad (6)$$

Proof. The initial projection $V^\dagger V$ is the support projection of $|C|$, while polar functional calculus gives $|C^\dagger|^{1/2} V = V|C|^{1/2}$. Consequently,

$$\begin{aligned} P^\dagger P &= |C^\dagger| \otimes \mathbf{1} + |C|^{1/2} V^\dagger V |C|^{1/2} \otimes B^\dagger B \\ &\quad - C \otimes B - C^\dagger \otimes B^\dagger \\ &= (|C| + |C^\dagger|) \otimes \mathbf{1} - C \otimes B - C^\dagger \otimes B^\dagger. \end{aligned}$$

This also shows explicitly why no invertibility assumption is needed. □

Lemma 2.2 (Sharp scalar bound). *For $|z| = 1$,*

$$\sum_{y=0}^{d-1} |1 + \omega^y z| \leq M_d, \quad (7)$$

with equality exactly when

$$z^d = (-1)^{d-1}. \quad (8)$$

Proof. Write $z = e^{2is}$, $h = \pi/d$, and $a = h/2$. Half the left side is

$$G_d(s) = \sum_{y=0}^{d-1} |\cos(s + yh)|,$$

which has period h .

If $d = 2m+1$, take $-a \leq s \leq a$ and reindex by $k = -m, \dots, m$. Every $s + kh$ lies in $[-\pi/2, \pi/2]$, since $a + mh = \pi/2$, and the finite cosine sum gives

$$G_d(s) = \sum_{k=-m}^m \cos(s + kh) = \csc(a) \cos s.$$

Its maximum occurs exactly at $s \equiv 0 \pmod{h}$, equivalently $z^d = 1$.

If $d = 2m$, take $0 \leq s \leq h$. The first m cosines are nonnegative and the last m nonpositive; endpoints cause no ambiguity because the boundary cosine vanishes. Two finite cosine sums give

$$G_d(s) = \sum_{y=0}^{m-1} \cos(s + yh) - \sum_{y=m}^{2m-1} \cos(s + yh) = \csc(a) \cos(s - a).$$

Its maximum occurs exactly at $s \equiv a \pmod{h}$, equivalently $z^d = -1$. Multiplying by two proves both claims. $\square$

Because $\mathbf{1} + \omega^y U$ is normal,

$$|C_y| = |\mathbf{1} + \omega^y U|, \quad |C_y^\dagger| = A_0 |\mathbf{1} + \omega^y U| A_0^\dagger.$$

Functional calculus and Lemma 2.2 give $F_d(U) \leq M_d \mathbf{1}$. Summing Lemma 2.1 therefore yields the exact gap decomposition

$$\boxed{M_d \mathbf{1} - \mathcal{I}_d = \frac{1}{2} \sum_{y=0}^{d-1} P_y^\dagger P_y + \frac{1}{2} \left[M_d \mathbf{1} - F_d(U) + A_0 (M_d \mathbf{1} - F_d(U)) A_0^\dagger \right] \otimes \mathbf{1}.} \quad (9)$$

Every term is positive. This proves $\mathcal{I}_d \leq M_d \mathbf{1}$ for arbitrary unitaries, even without the order relations, and (4) follows from $\text{Re}(A_0 \otimes B_d) \leq \mathbf{1}$.

3 Weighted-shift maximizers

Let

$$X|j\rangle = |j+1 \bmod d\rangle, \quad \delta_d = \begin{cases} 0, & d \text{ odd}, \\ 1, & d \text{ even}, \end{cases}$$

and enumerate the equality roots from Lemma 2.2 by

$$z_k = \exp\left(\frac{\pi i(2k + \delta_d)}{d}\right), \quad k = 0, \dots, d-1. \quad (10)$$

For any ordering $\kappa = (\kappa_0, \dots, \kappa_{d-1})$ of these roots, define

$$\begin{aligned} Z_\kappa &= \text{diag}(z_{\kappa_0}, \dots, z_{\kappa_{d-1}}), \\ s_{y,k} &= \frac{1 + \omega^y z_k}{|1 + \omega^y z_k|}, \\ S_y^\kappa &= \text{diag}(s_{y,\kappa_0}, \dots, s_{y,\kappa_{d-1}}), \\ H_y^\kappa &= \text{diag}(|1 + \omega^y z_{\kappa_0}|, \dots, |1 + \omega^y z_{\kappa_{d-1}}|). \end{aligned}$$

The denominators never vanish because -1 is not a root of $t^d - (-1)^{d-1}$.

Proposition 3.1 (Every root ordering gives a maximizer). *On $\mathbb{C}^d \otimes \mathbb{C}^d$, set*

$$\begin{aligned} A_0 &= X, & A_1 &= XZ_\kappa, \\ V_y &= XS_y^\kappa, & B_y &= \overline{V_y} \quad (0 \leq y < d), \\ B_d &= X, \end{aligned} \quad (11)$$

and use

$$|\Phi_d\rangle = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} |j, j\rangle.$$

All observables are unitary, all have d -th power $\mathbf{1}$, and

$$\langle \Phi_d | \overline{\mathcal{I}}_d | \Phi_d \rangle = M_d + 1.$$

Proof. The z_k are all roots of $t^d - (-1)^{d-1}$. Comparing the constant term and evaluating the polynomial at -1 give

$$\prod_k z_k = 1, \quad \prod_k (1 + \omega^y z_k) = 2. \quad (12)$$

Multiplication by ω^y merely permutes the roots. The second product is real and positive, so

$$\prod_k s_{y,k} = 1.$$

For arbitrary weights t_j ,

$$(X \text{diag}(t_0, \dots, t_{d-1}))^d = \left(\prod_j t_j \right) \mathbf{1}. \quad (13)$$

Equations (12)–(13) prove every order relation in (11); unitarity is immediate.

Moreover,

$$A_0 + \omega^y A_1 = X(\mathbf{1} + \omega^y Z_\kappa) = V_y H_y^\kappa,$$

which is its polar decomposition. The maximally entangled trace identity

$$\langle \Phi_d | R \otimes S | \Phi_d \rangle = \frac{1}{d} \text{Tr}(RS^T)$$

and $B_y^T = V_y^\dagger$ give

$$\langle \mathcal{I}_d \rangle = \frac{1}{d} \sum_{j,y} |1 + \omega^y z_{\kappa_j}| = \frac{1}{d} \sum_j M_d = M_d.$$

Finally, $X \otimes X$ fixes $|\Phi_d\rangle$, so the added correlator equals one. The upper bound (4) proves global optimality. $\square$

4 The designated output distribution

The eigenvectors of $A_0 = X$ and $A_1 = XZ_\kappa$ can be written

$$u_c = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} \omega^{-cj} |j\rangle, \quad v_a = \frac{1}{\sqrt{d}} \sum_{j=0}^{d-1} q_j \omega^{-aj} |j\rangle,$$

where

$$q_0 = 1, \quad q_{j+1} = z_{\kappa_j} q_j. \quad (14)$$

The sequence is cyclic because $\prod_k z_k = 1$. Put

$$\hat{q}_m = \sum_{j=0}^{d-1} q_j \omega^{mj}.$$

The transpose of the outcome- b projector of $B_d = X$ is the outcome- $(-b)$ projector of A_0 . Hence

$$p(a, b \mid 1, d) = \frac{1}{d} |\langle u_{-b}, v_a \rangle|^2 = \frac{|\hat{q}_{-(a+b)}|^2}{d^3}. \quad (15)$$

Parseval gives $\sum_m |\hat{q}_m|^2 = d^2$. As $a + b$ traverses all residues when either output is fixed, (15) immediately implies

$$\sum_b p(a, b \mid 1, d) = \sum_a p(a, b \mid 1, d) = \frac{1}{d}. \quad (16)$$

For $d \geq 4$, choose

$$\kappa = (0, 1, \dots, d-3, d-1, d-2), \quad (17)$$

the canonical ordering with its final two entries exchanged. Consider the second cyclic autocorrelation

$$C_2 = \sum_j q_{j+2} \overline{q_j} = \sum_j z_{\kappa_j} z_{\kappa_{j+1}}.$$

For the canonical ordering this sum vanishes. Exchanging the last two entries changes only the affected summands and gives

$$C_2 = (z_{d-1} - z_{d-2})(z_{d-3} - z_0), \quad |C_2| = 4 \sin(\pi/d) \sin(3\pi/d) > 0. \quad (18)$$

Thus the Fourier magnitudes cannot all equal d , so the joint table is not uniform.

For the quantitative gap, put

$$x_m = |\widehat{q}_m|^2 - d, \quad \Delta = \max_m x_m.$$

Then $\sum_m x_m = 0$, and Fourier inversion at lag two gives

$$C_2 = \frac{1}{d} \sum_m x_m \omega^{-2m}.$$

Here the subtracted constant contributes zero because $d \geq 4$ and $\sum_m \omega^{-2m} = 0$. The nonzero mean-zero sequence x_m has at most $d - 1$ positive entries. Therefore

$$\sum_m |x_m| = 2 \sum_{x_m > 0} x_m \leq 2(d - 1)\Delta,$$

and

$$\Delta \geq \frac{d|C_2|}{2(d - 1)}.$$

With trivial Eve, the optimal guessing strategy selects a most likely output pair, so

$$G = \max_{a,b} p(a, b \mid 1, d) = \frac{d + \Delta}{d^3}.$$

Substitution of (18) proves (5) and Theorem 1.1.

5 A sparse exact four-outcome certificate

Let $\zeta = e^{\pi i/8}$, so $\mathbb{Q}(\zeta) \cong \mathbb{Q}[x]/(x^8 + 1)$. At $d = 4$, choose $\kappa = (0, 1, 3, 2)$. Proposition 3.1 becomes

$$A_0 = X, \quad A_1 = X \operatorname{diag}(\zeta^2, \zeta^6, \zeta^{14}, \zeta^{10}), \quad B_4 = X.$$

The four polar unitaries $V_y = X \operatorname{diag}(\zeta^{e_{y,0}}, \dots, \zeta^{e_{y,3}})$ are encoded by

y	$e_{y,0}$	$e_{y,1}$	$e_{y,2}$	$e_{y,3}$	
0	1	3	15	13	$B_y = \overline{V_y}.$
1	3	13	1	15	
2	13	15	3	1	
3	15	1	13	3	

These sparse monomial formulas are exact matrices over $\mathbb{Q}(\zeta_{16})$; no numerical polar decomposition is part of the certificate.

The sequence (14) is

$$q = (1, \zeta^2, -1, \zeta^6).$$

With $\omega = i$, direct finite Fourier transformation gives

$$\widehat{q} = (i\sqrt{2}, 2 + i\sqrt{2}, -i\sqrt{2}, 2 - i\sqrt{2}), \quad (|\widehat{q}_m|^2)_m = (2, 6, 2, 6).$$

Equation (15) therefore gives

$$p(a, b \mid 1, 4) = \begin{cases} 1/32, & a + b \text{ even,} \\ 3/32, & a + b \text{ odd.} \end{cases} \quad (19)$$

Consequently,

$$G(AB \mid 1, 4, E) = \frac{3}{32} > \frac{1}{16}, \quad H_{\min}(AB \mid E) = 5 - \log_2 3 < 4.$$

The independent verifier reconstructs the observables from the displayed exponents and checks, using exact rational arithmetic modulo $x^8 + 1$, all unitarity and fourth-order relations, all polar factorizations, the original Bell expression, both probability calculations, uniform marginals, and the strict guessing gap.

6 Consequences, verification, and scope

1. The maximal-randomness assertion in Conjecture 2 of Ref. [1], interpreted for its displayed first augmented Bell family, is false for every $d \geq 4$.
2. The obstruction needs neither a direct sum nor an Eve-held flag. It occurs on one $d \times d$ block, on a maximally entangled state, while both local marginals remain uniform.
3. The Bell maximum does not determine a unique target behavior in these dimensions. In particular, the first augmented family cannot self-test a unique maximizing correlation.
4. No universal near-maximal bound converging to $1/d^2$ as the Bell deficit tends to zero can hold for this family when $d \geq 4$, because the exact maximizing face already contains nonuniform behaviors.
5. The complete maximizing face, its worst guessing probability, and the maximal-violation randomness question for $d = 2, 3$ remain unresolved here. The Appendix proves only a multiplicity restriction, not a classification.

The source preprint’s Appendix B.1 fixes the *full canonical probability distribution* before running its NPA guessing-probability calculation. That calculation can support randomness of the fixed canonical behavior without excluding another behavior at the same Bell maximum. The strategy above supplies exactly such a behavior. This note does not dispute the separate protocols of Farkas, Mironowicz, and Augusiak [3], nor the distinct three-outcome robustness study of D’Avino et al. [4].

The accompanying package contains the TeX source, the exact $\mathbb{Q}(\zeta_{16})$ verifier, a compact machine-readable certificate, the all-dimensional constructor and symbolic family certificate, floating point regressions for $d = 2, \dots, 12$, historical discovery code, claims and assumptions ledgers, failed approaches, a research log, and SHA-256 hashes. The finite programs support the proof but do not formally quantify over all d .

A targeted search on 26 July 2026 found no public equivalent of the weighted-shift counterexample. This does not establish absolute novelty: unindexed, unpublished, or concurrent work may exist. The appropriate priority wording is therefore “to the best of our knowledge.” No originating author or other researcher was contacted.

The proof architecture, derivations, manuscript, verification code, and literature audit were developed with heavy assistance from ChatGPT 5.6 Sol under Alec Kriebel's direction. Alec Kriebel is the named human author but cannot independently validate the mathematical claims. Independent expert checking remains necessary.

A A support-level multiplicity restriction

This appendix records a structural consequence that is not needed for the counterexample.

Lemma A.1 (Reflection-product rank bound). *Let T be a unitary on an n -dimensional space and let E be an orthogonal projection of rank r . If*

$$T^d = \mathbf{1}, \quad (T(\mathbf{1} - 2E))^d = -\mathbf{1},$$

then $n \leq dr$.

Proof. Expanding the second product and using $T^d = \mathbf{1}$ expresses $-\mathbf{1}$ as a product of d conjugates of the reflection $\mathbf{1} - 2E$. For arbitrary operators $R_1, \dots, R_d$,

$$\text{rank}(\mathbf{1} - R_1 \cdots R_d) \leq \sum_{j=1}^d \text{rank}(\mathbf{1} - R_j),$$

by repeatedly using $\mathbf{1} - AB = (\mathbf{1} - A) + A(\mathbf{1} - B)$. Each reflected defect has rank r ; the left side is $\text{rank}(\mathbf{1}) = n$. $\square$

Proposition A.2 (Equal supported multiplicities). *Let a finite-dimensional admissible strategy and state $|\Psi\rangle_{ABE}$ attain $\langle \bar{\mathcal{I}}_d \rangle = M_d + 1$. Put*

$$\rho_A = \text{Tr}_{BE} |\Psi\rangle\langle\Psi|, \quad K = \text{supp } \rho_A, \quad U = A_0^\dagger A_1.$$

On K , all d roots of $z^d = (-1)^{d-1}$ occur as eigenvalues of U with the same multiplicity. In particular,

$$d \mid \dim K.$$

No statement is made about unused sectors orthogonal to K .

Proof. Equality in the two bounds $\langle \mathcal{I}_d \rangle \leq M_d$ and $\langle \text{Re}(A_0 \otimes B_d) \rangle \leq 1$ is forced separately. Every positive term in (9) therefore vanishes on the state:

$$P_y |\Psi\rangle = 0, \tag{20}$$

$$(M_d \mathbf{1} - F_d(U)) \otimes \mathbf{1} |\Psi\rangle = 0, \tag{21}$$

$$(A_0 \otimes B_d \otimes \mathbf{1}_E) |\Psi\rangle = |\Psi\rangle. \tag{22}$$

The last implication uses $\mathbf{1} - \text{Re } W = \frac{1}{2}(\mathbf{1} - W)^\dagger(\mathbf{1} - W)$ for unitary W .

For any Alice operator R ,

$$(R \otimes \mathbf{1}) |\Psi\rangle = 0 \iff R|_K = 0, \tag{23}$$

as follows from a Schmidt decomposition across $A : (BE)$. Thus (21) places K inside the equality spectral subspace

$$L = \text{ran } \mathbf{1}_{\mathcal{Z}_d}(U), \quad \mathcal{Z}_d = \{z : |z| = 1, z^d = (-1)^{d-1}\}.$$

Equation (22) gives $A_0K = K$.

We next justify a cancellation in (20) without assuming that a polar factor is globally invertible. Let $C_y = V_y|C_y|$ be the canonical polar decomposition, let

$$E_y^{\text{bad}} = \mathbf{1}_{\{-\omega^{-y}\}}(U), \quad F_y = \text{supp}|C_y^\dagger| = A_0(\mathbf{1} - E_y^{\text{bad}})A_0^\dagger.$$

The bad phase is not in $\mathcal{Z}_d$, so $E_y^{\text{bad}}K = 0$. Together with $A_0K = K$, this gives $F_y|_K = \mathbf{1}_K$. Also $F_yV_y = V_y$, because F_y is the final support of the polar partial isometry. Since

$$P_y = (|C_y^\dagger|^{1/2} \otimes \mathbf{1})(\mathbf{1} - V_y \otimes B_y),$$

the vector

$$q_y = (\mathbf{1} - V_y \otimes B_y \otimes \mathbf{1}_E)|\Psi\rangle$$

belongs to $\text{ran } F_y \otimes \mathcal{H}_{BE}$. Equation (20) places the same vector in the kernel of $|C_y^\dagger|^{1/2} \otimes \mathbf{1}$, which is orthogonal to that range. Hence $q_y = 0$, or

$$(V_y \otimes B_y \otimes \mathbf{1}_E)|\Psi\rangle = |\Psi\rangle. \quad (24)$$

The space K lies in the initial support of V_y , so the polar factor is an isometry there. Tracing (24) over BE gives $\rho_A = V_y\rho_A V_y^\dagger$, and therefore $V_yK = K$. Hence

$$S_y = A_0^\dagger V_y$$

preserves K . On L , functional calculus gives

$$S_y = s_y(U), \quad s_y(z) = \frac{1 + \omega^y z}{|1 + \omega^y z|}, \quad s_y(z)^2 = \omega^y z.$$

It follows that $U = \omega^{-y}S_y^2$ preserves K . We may now let E_k denote the spectral projections of $U|_K$ for the d roots in $\mathcal{Z}_d$.

Put $\eta = e^{\pi i/d}$, and read $y+1$ modulo d , so $S_d = S_0$ and $V_d = V_0$ in the next relation. Comparing adjacent half-angle phases shows that for each y exactly one equality root crosses the sign cut:

$$\frac{1+w}{|1+w|} = e^{i\theta/2} \quad \text{when } w = e^{i\theta}, \quad -\pi < \theta < \pi.$$

Multiplication by ω advances the half-angle by π/d ; among the equally spaced equality roots, exactly one principal argument wraps across π , contributing an additional minus sign. Thus

$$S_y^\dagger S_{y+1} = \eta(\mathbf{1} - 2E_{k(y)}), \quad V_{y+1} = \eta V_y(\mathbf{1} - 2E_{k(y)}), \quad (25)$$

and $y \mapsto k(y)$ is a permutation of the d root labels. Raising (24) to the d -th power, using $B_y^d = \mathbf{1}$, and applying (23) gives $V_y^d = \mathbf{1}$ on K . Since $\eta^d = -1$, equation (25) gives

$$(V_y(\mathbf{1} - 2E_{k(y)}))^d = -\mathbf{1}.$$

Lemma A.1, with $n = \dim K$, yields

$$n \leq d \text{ rank } E_k$$

for every k . But the E_k sum to $\mathbf{1}_K$, so their ranks sum to n . Every inequality must therefore be an equality: $\text{rank } E_k = n/d$ for all k . $\square$

Remark A.3. Proposition A.2 is restricted to exact maximizers of the *augmented* functional because the proof uses the A_0 - B_d stabilizer. Equal multiplicities do not imply a Weyl representation, uniqueness, or self-testing; the weighted-shift family is the obstruction.

References

- [1] I. Perito, R. D’Avino, M. Jung, P. Mironowicz, A. Acín, and R. Augusiak, Bell inequalities tailored to optimal global randomness certification, *arXiv:2606.21362v3* (2026), <https://arxiv.org/abs/2606.21362>.
- [2] A. Kriebel, The exact quantum value of a cyclic Bell operator, provisional AI-assisted research note (2026), [public paper page](#).
- [3] M. Farkas, P. Mironowicz, and R. Augusiak, Maximal global device-independent randomness from projective measurements in every dimension, *arXiv:2606.21369v2* (2026), <https://arxiv.org/abs/2606.21369>.
- [4] R. D’Avino, I. Perito, P. Mironowicz, A. Acín, and R. Augusiak, Noise robustness of three outcome Bell certified quantum randomness, *arXiv:2606.21371v2* (2026), <https://arxiv.org/abs/2606.21371>.